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(54) **SYSTEMS AND METHODS FOR IMPROVING CALL CENTER FEATURES USING GENERATIVE AI**

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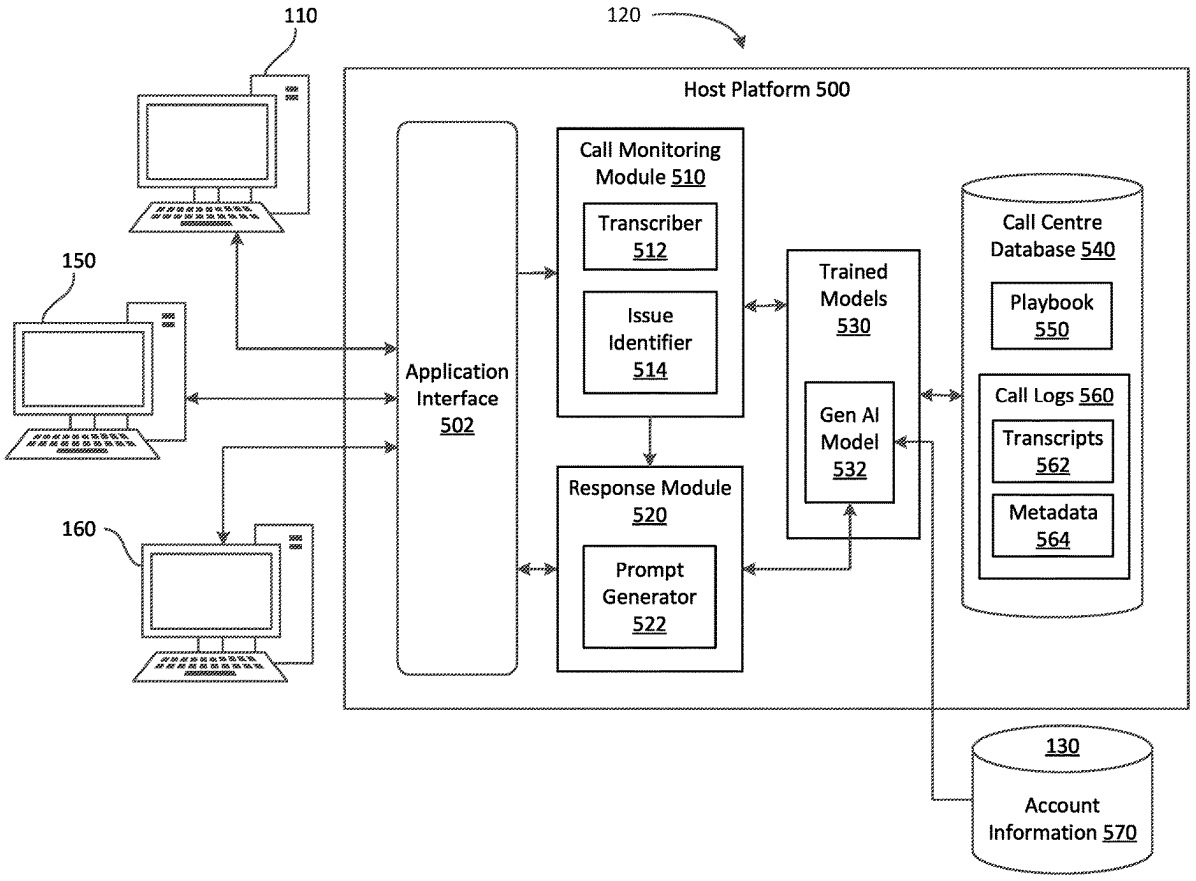
(57) **ABSTRACT**

The present disclosure relates to systems and methods for enhancing call center features using generative AI. There is provided a server computer system, comprising: a processor, a communications module coupled to the processor, and a memory coupled to the processor. The memory stores a playbook of a call center and instructions that, when executed, configure the processor to monitor a call in real-time during the call with a caller, identify, from the call, a caller issue in real-time using a trained machine learning model, obtain a response to the caller issue based on execution of a generative artificial intelligence (GenAI) model and the playbook stored in the memory, and implement the response during the call.



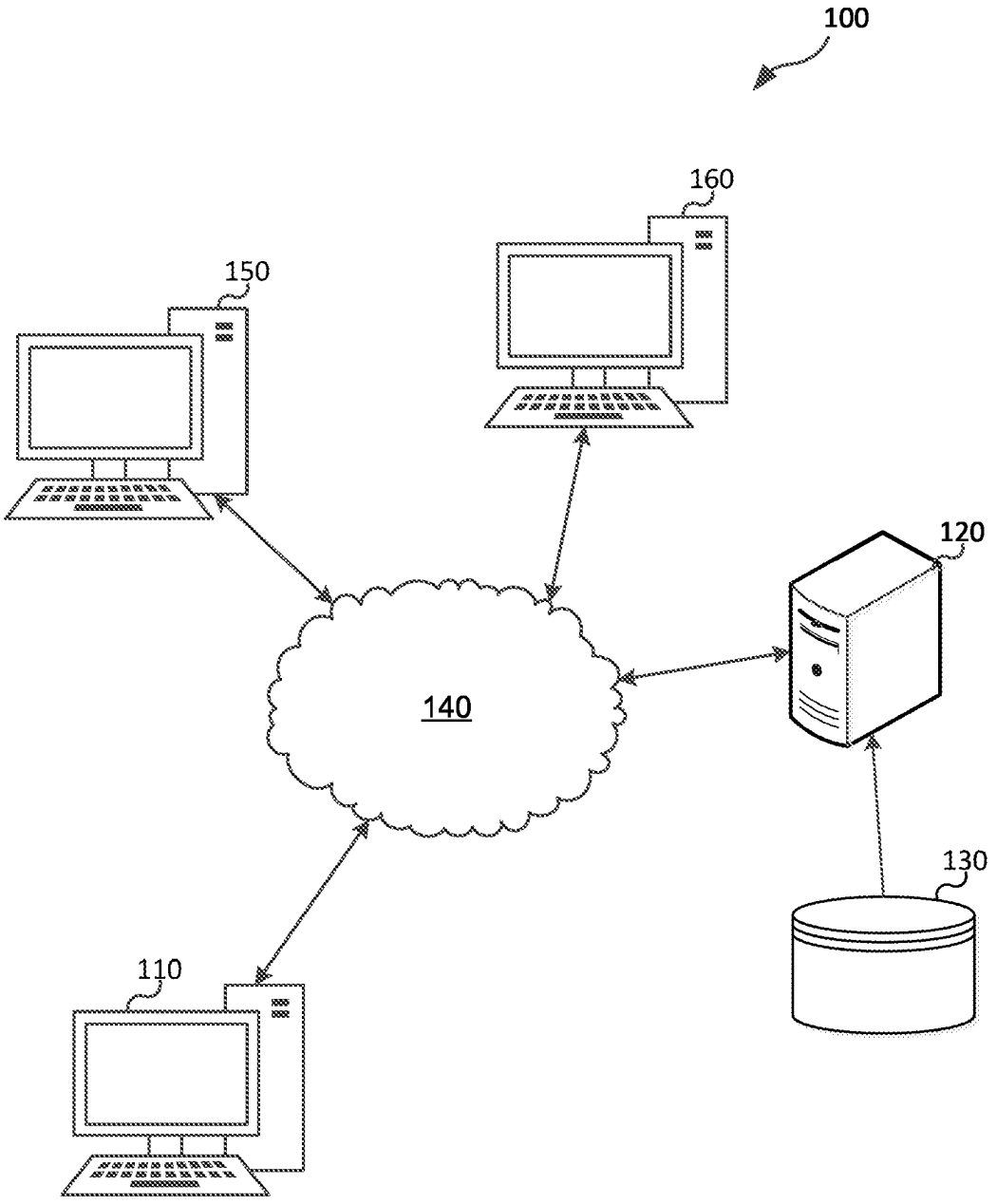


FIG. 1

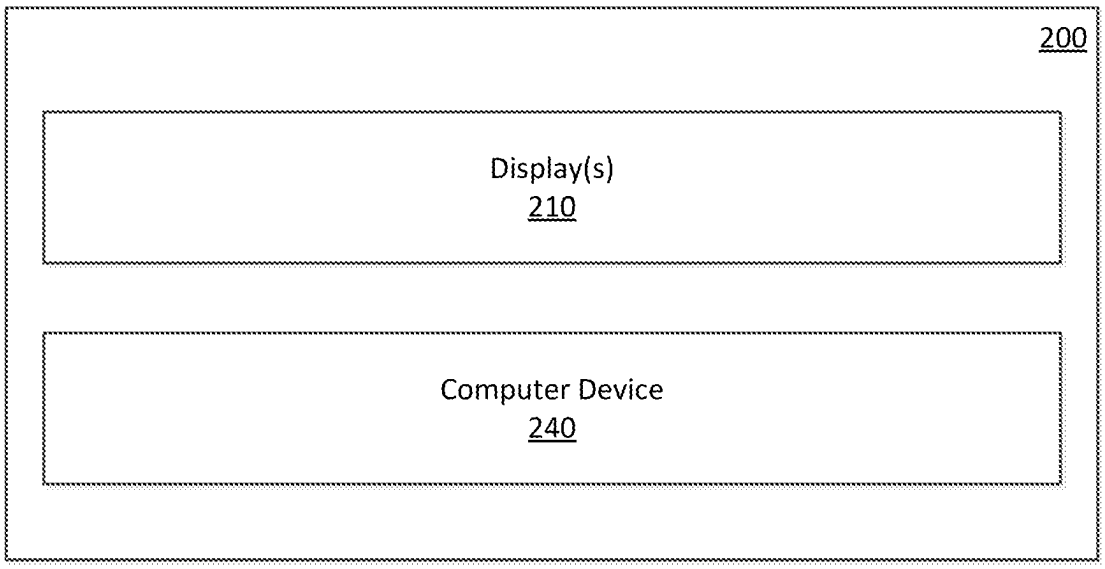


FIG. 2

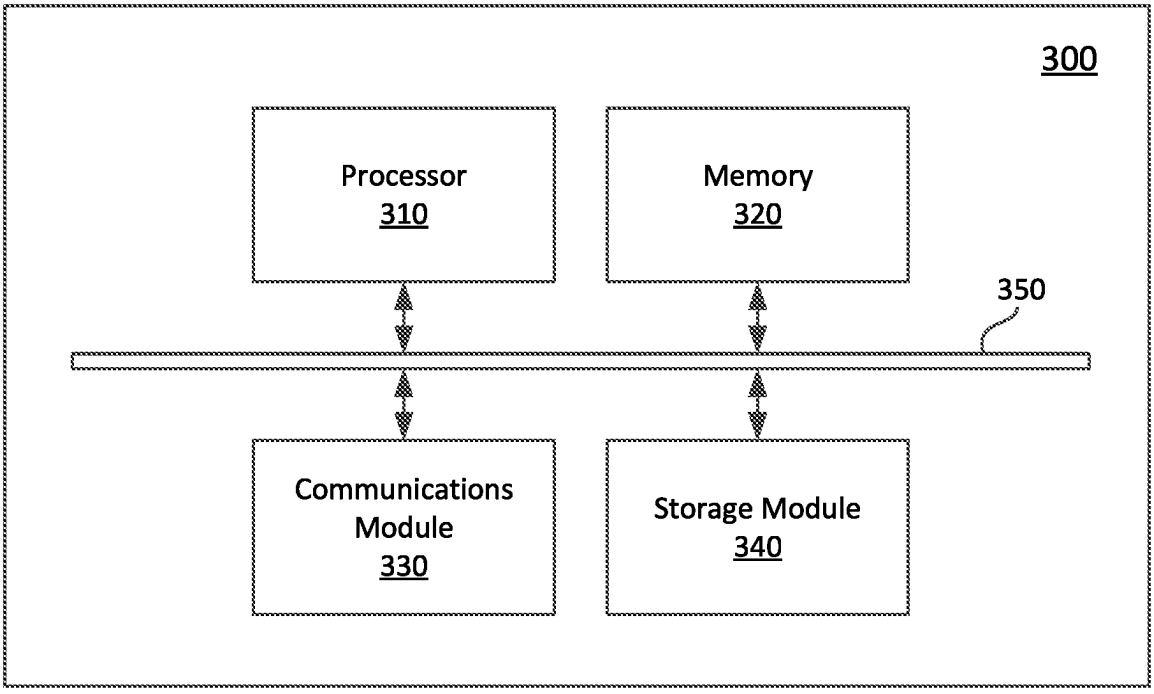

300
Processor
310
Memory
320
350
Communications
Module
330
Storage Module
340

FIG. 3

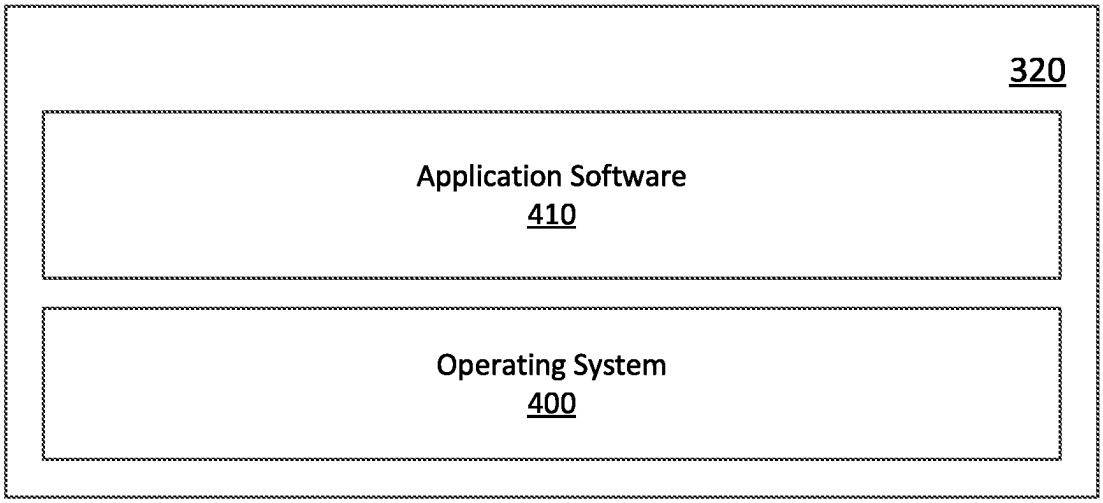

320
Application Software
410
Operating System
400

FIG. 4

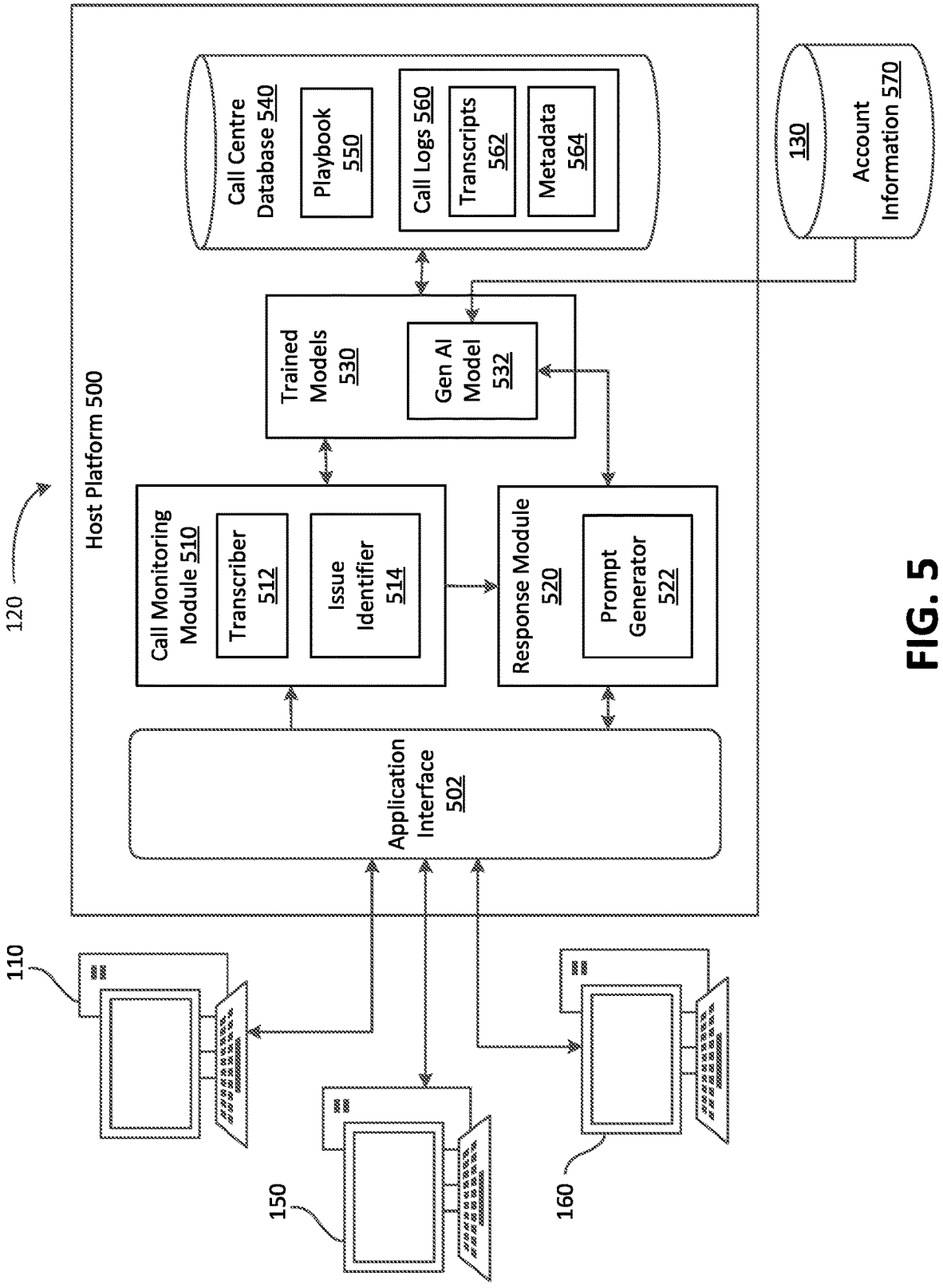

120
Host Platform 500
Call Centre Database 540
Playbook 550
Call Logs 560
Transcripts 562
Metadata 564
130
Account Information 570
Trained Models 530
Gen AI Model 532
Call Monitoring Module 510
Transcriber 512
Issue Identifier 514
Response Module 520
Prompt Generator 522
Application Interface 502
110
150
160

FIG. 5

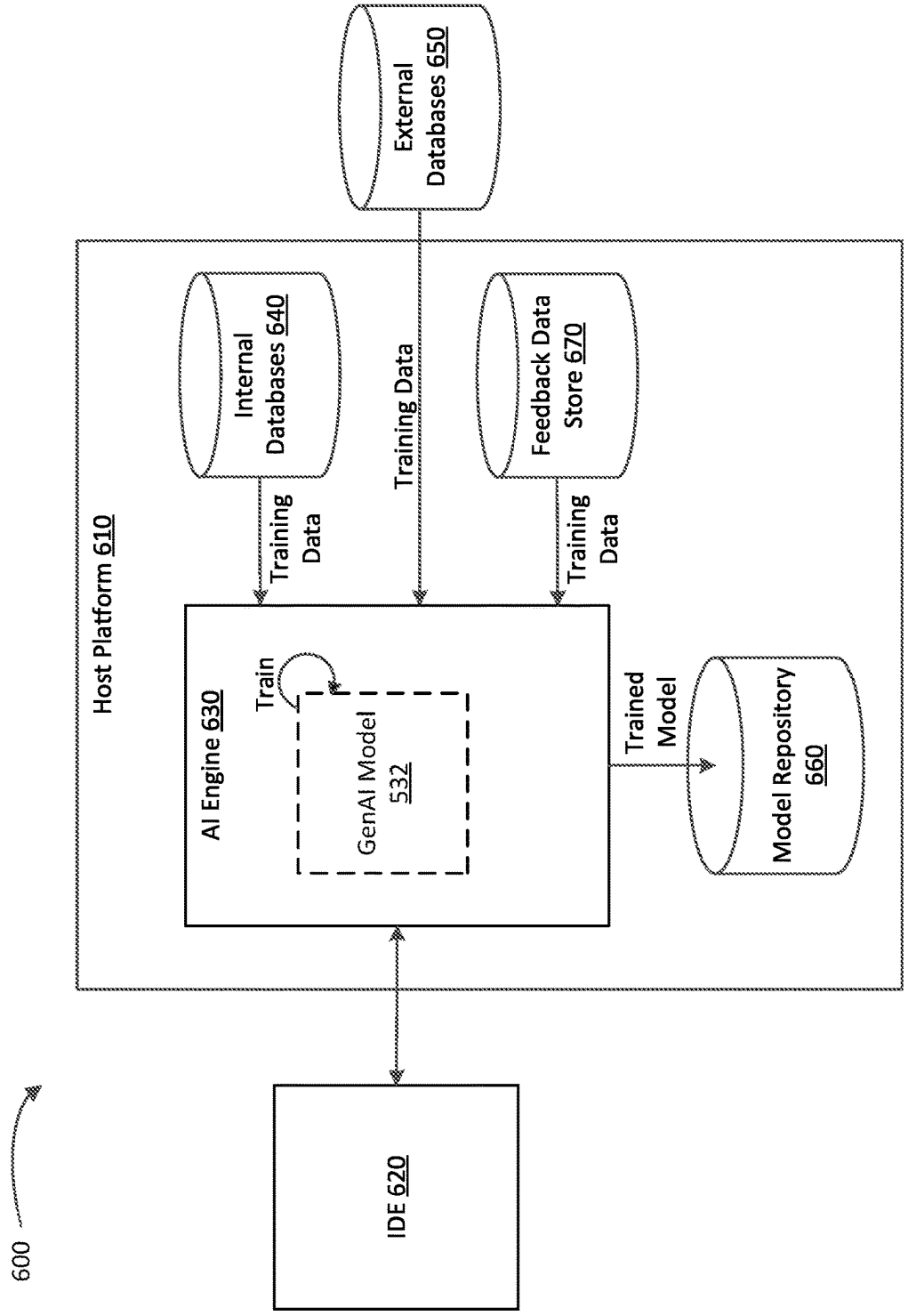

600
IDE 620
Host Platform 610
External Databases 650
Internal Databases 640
Training Data
Feedback Data Store 670
Training Data
Training Data
AI Engine 630
Train
GenAI Model 532
Trained Model
Model Repository 660

FIG. 6

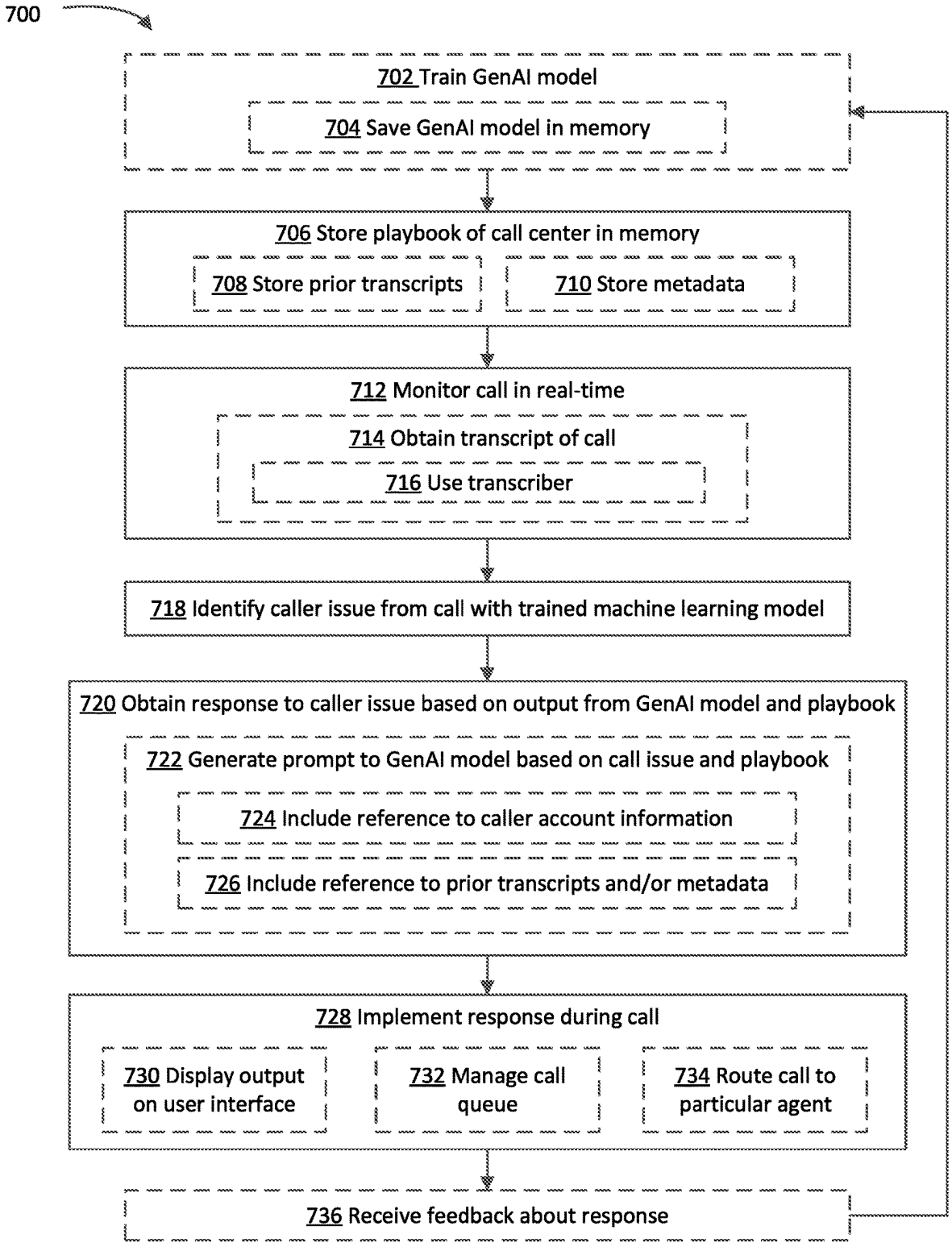

700
702 Train GenAI model
704 Save GenAI model in memory
706 Store playbook of call center in memory
708 Store prior transcripts
710 Store metadata
712 Monitor call in real-time
714 Obtain transcript of call
716 Use transcriber
718 Identify caller issue from call with trained machine learning model
720 Obtain response to caller issue based on output from GenAI model and playbook
722 Generate prompt to GenAI model based on call issue and playbook
724 Include reference to caller account information
726 Include reference to prior transcripts and/or metadata
728 Implement response during call
730 Display output on user interface
732 Manage call queue
734 Route call to particular agent
736 Receive feedback about response

FIG. 7

SYSTEMS AND METHODS FOR IMPROVING CALL CENTER FEATURES USING GENERATIVE AI

TECHNICAL FIELD

[0001] The present application relates to call centers and, more particularly, to systems and methods for improving call center features using generative AI.

BACKGROUND

[0002] Ideally, when calling a call center, a customer would be immediately connected to an agent with complete information and the requisite experience to provide direct assistance to the caller. However, that is often not possible or difficult, as call center playbooks can be long and complicated, there may be more callers than available agents, the agent may not have prior experience with the particular issue(s) of the caller, and/or the agent may not have all the requisite information immediately on hand. This may delay the caller from receiving their desired assistance in a timely fashion.

[0003] Meanwhile, generative artificial intelligence (GenAI) is an artificial intelligence that is capable of generating text, images, and other media from generative artificial intelligence models such as large language models, multimodal large language models, neural networks, and the like. A GenAI model can learn patterns and structure of the training data input to the GenAI model during training, and then use what is learned during the training to generate new data with similar characteristics.

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] Embodiments are described in detail below, with reference to the following drawings:

[0005] FIG. **1** is a schematic operations diagram illustrating an operating environment of a system according to an example embodiment of the present disclosure;

[0006] FIG. **2** is a simplified schematic diagram showing components of an example computer device;

[0007] FIG. **3** is a high-level schematic diagram of an example computer system;

[0008] FIG. **4** shows a simplified organization of software components stored in a memory of the computer system of FIG. **3**; and

[0009] FIG. **5** is a schematic diagram illustrating a generative artificial intelligence (GenAI) computing environment of the server computer system of FIG. **1** according to example embodiments;

[0010] FIG. **6** is a diagram illustrating processes for training a machine learning model according to example embodiments; and

[0011] FIG. **7** is a flowchart showing operations performed by the server computer system of FIG. **5** for enhancing call center features according to example embodiments.

[0012] Like reference numerals are used in the drawings to denote like elements and features.

DETAILED DESCRIPTION

[0013] In one aspect of the present disclosure, there is provided a server computer system, comprising: a processor; a communications module coupled to the processor; and a memory coupled to the processor, the memory storing a playbook of a call center and instructions that, when executed, configure the processor to: monitor a call in real-time during the call with a caller; identify, from the call, a caller issue in real-time using a trained machine learning model; obtain a response to the caller issue based on execution of a generative artificial intelligence (GenAI) model and the playbook stored in the memory; and implement the response during the call.

[0014] In some implementations, the instructions further configure the processor to obtain a transcript of the call in real-time during the call, and identify the caller issue from the transcript of the call.

[0015] In some implementations, the server computer system further comprises a transcriber coupled to the processor, wherein the transcript of the call is obtained by the transcriber.

[0016] In some implementations, the instructions further configure the processor to train the GenAI model with other call center playbooks prior to the call.

[0017] In some implementations, the GenAI model is a large language model (LLM).

[0018] In some implementations, the instructions further configure the processor to obtain the response by: generating a prompt to the LLM, the prompt including the caller issue with reference to the playbook; and obtain an output from the LLM responsive to the prompt; and implement the response by providing the output via a user interface.

[0019] In some implementations, the output comprises a summary of a portion of the playbook related to the caller issue, and the prompt to the LLM is for generating the summary as the output.

[0020] In some implementations, the instructions further configure the processor to receive an authorization regarding an identity of the caller from an agent device.

[0021] In some implementations, the prompt further includes reference to account information associated with the identity of the caller.

[0022] In some implementations, the caller issue is a form of a question, the output comprises a reply to the question, and the prompt to the LLM is for generating the reply based on the playbook customized according to the account information.

[0023] In some implementations, the memory further stores prior call transcripts of the call center, wherein the prompt to the LLM is for identifying: the caller issue in one of the prior call transcripts, and a solution to the caller issue in the one of the prior call transcripts; and wherein the output comprises the solution.

[0024] In some implementations, the instructions further configure the processor to receive feedback about the solution via the user interface, retrain the GenAI model based on execution of the GenAI model on the caller issue, the solution, and the feedback, and store the retrained GenAI model in the memory.

[0025] In some implementations, the memory further stores call log data of the call center, the call log data comprising prior call transcripts of prior calls and metadata associated with the prior calls, wherein the instructions, when executed, further configure the processor to: obtain the response by: generating a prompt to the GenAI model, the prompt including the caller issue with reference to the playbook and the call log data, and obtain an output from the GenAI model responsive to the prompt; and implement the response based on the output.

[0026] In some implementations, the metadata comprises a call duration of each of the prior calls, wherein the prompt to the GenAI model is for identifying: the caller issue in one of the prior call transcripts, and the call duration of the prior call of the one of the prior call transcripts; wherein the output comprises an expected call duration of the call based on the call duration of the prior call of the one of the prior call transcripts; and wherein the instructions, when executed, further configure the processor to implement the response by placing the call in a call queue according to the expected call duration.

[0027] In some implementations, the prompt further includes account information associated with an identity of the caller, and wherein the expected call duration is further based on the account information of the caller.

[0028] In some implementations, the instructions further configure the processor to implement the response by routing the call to a particular agent based on the output.

[0029] In some implementations, the metadata comprises identification of an agent associated each of the prior calls, wherein the prompt to the GenAI model is for identifying: the caller issue in one of the prior call transcripts, and the identification of the agent associated with the prior call of the one of the prior call transcripts; wherein the output comprises the identification of the agent associated with the prior call of the one of the prior call transcripts; and wherein the instructions, when executed, further configure the processor to implement the response by routing the call to the identified agent.

[0030] In some implementations, the instructions further configure the processor to receive feedback about the routing from the identified agent, retrain the GenAI model based on execution of the GenAI model on the caller issue, the routing, and the feedback, and store the retrained GenAI model in the memory.

[0031] In another aspect of the present disclosure, there is provided a method comprising: storing a playbook of a call center in a memory; monitoring a call in real-time during the call with a caller; identifying, from the call, a caller issue in real-time using a trained machine learning model; obtaining a response to the caller issue based on execution of a generative artificial intelligence (GenAI) model and the playbook stored in the memory; and implementing the response during the call.

[0032] In a further aspect of the present disclosure, there is provided a computer-readable medium comprising instructions stored therein which, when executed by a processor, cause a computer to: store a playbook of a call center in a memory; monitor a call in real-time during the call with a caller; identify, from the call, a caller issue in real-time using a trained machine learning model; obtain a response to the caller issue based on execution of a generative artificial intelligence (GenAI) model and the playbook stored in the memory; and implement the response during the call.

[0033] The present subject matter uses trained machine learning models and generative AI to identify issues in customer calls, identify the relevant information within a repository of the call center's playbook, provide relevant playbook summaries, answers to queries customized to the caller, identify solutions to the caller issue within past call transcripts, and manage the call center queue, route calls, and the like. The models can be used to help provide more accurate and efficient responses to callers than is normally performed within a call center, and to reduce the computing processing resources otherwise required.

[0034] FIG. **1** is a schematic operation diagram illustrating an operating environment of an example embodiment of a contact or call center. As shown, the system **100** includes an agent device **110** and a server computer system **120** with a database **130**, coupled to one another through a network **140**, which may include a public network such as the Internet and/or a private network. The agent device **110** and the server computer system **120** may be in the same location or geographically disparate locations. In other words, the agent device **110** and the server computer system **120** may be located remote from one another. The system **100** may further include other agent devices **150** and **160**, which may also be coupled the server computer system **120** through the network **140**.

[0035] The agent device **110** may be a personal computer as shown in FIG. **1**. However, the agent device **110** may be a computing device of another type such as for example a smartphone, a laptop, a tablet computer, a notebook computer, a hand-held computer, a personal digital assistant, a portable navigation device, a mobile phone, a wearable computing device (e.g., a smart watch, a wearable activity monitor, wearable smart jewelry, and glasses and other optical devices that include optical head-mounted displays), an embedded computing device (e.g., in communication with a smart textile or electronic fabric), and any other type of computing device that may be configured to store data and software instructions, and execute software instructions to perform operations consistent with disclosed embodiments. The agent device **110** may be associated with a user, such as a call agent of the call center.

[0036] The server computer system **120** may be, for example, a mainframe computer, a minicomputer, or the like. In some embodiments thereof, a computer system may be formed of or may include one or more computing devices. The server computer system **120** may include and/or may communicate with multiple computing devices such as, for example, one or more database servers (including a database **130**), computer servers, and the like. Multiple computing devices such as these may be in communication using a computer network and may communicate to act in cooperation as a computer server system. For example, the computing devices may communicate using a local-area network (LAN). In some embodiments, the server computer system **120** may include multiple computing devices organized in a tiered arrangement. For example, the server computer system **120** may include middle tier and back-end computing devices. In some embodiments, the server computer system **120** may be a cluster formed of a plurality of interoperating computing devices.

[0037] The server computer system **120** may be associated with a call center used by one of various companies or institutions. In some embodiments, the server computer system **120** may be associated with a call center of a financial institution and, to that end, may maintain records of customer financial accounts and associated financial data in the database **130**. The database **130** may be provided internally within the server computer system **120** or externally. To that end, the database **130** may be provided remotely from the server computer system **120**. For example, the database **130** may be stored in one or more data centers, and the data centers may store data with bank-grade security.

[0038] The network **140** is a computer network. In some embodiments, the network **140** may be an internetwork such as may be formed of one or more interconnected computer networks. For example, the network **140** may be or may include an Ethernet network, an asynchronous transfer mode (ATM) network, a wireless network, a telecommunications network, or the like.

[0039] FIG. **1** illustrates an example representation of components of the system **100**. The system **100** can, however, be implemented differently than the example of FIG. **1**. For example, various components that are illustrated as separate systems in FIG. **1** may be implemented on a common system. By way of further example, the functions of a single component may be divided into multiple components. In another embodiment, the system **100** may be a cloud-based system. For example, the server computer system **120** may itself be virtual and the various components and modules thereof may be resident on the cloud. The server computer system **120** may include one or more virtual machines or virtual processors that may be accessed via the cloud.

[0040] FIG. **2** is a simplified schematic diagram showing components of an exemplary computing device **200**, such as the agent device **110**, **150**, **160**. The exemplary computing device **200** may include modules including, as illustrated, for example, one or more displays **210** and a computer device **240**.

[0041] The one or more displays **210** are a display module. The one or more displays **210** are used to display screens of a graphical user interface that may be used, for example, to communicate with the server computer system **120**. The one or more displays **210** may be internal displays of the exemplary computing device **200** (e.g., disposed within a body of the computing device).

[0042] The computer device **240** is in communication with the one or more displays **210**. The computer device **240** may be or may include a processor which is coupled to the one or more displays **210**.

[0043] Referring now to FIG. **3**, a high-level operation diagram of an example computer system **300** is shown. In some embodiments, the example computing system **300** may be exemplary of the server computer system **120** and/or the agent devices **110**, **150**, **160** (shown in FIG. **1**). The example computer system **300** includes a variety of modules. For example, the example computer system **300** may include at least one processor **310**, a memory **320**, a communications module **330**, and/or a storage module **340**. As illustrated, the foregoing example modules of the example computer system **300** are in communication over a bus **350**.

[0044] The at least one processor **310** is a hardware processor. The at least one processor **310** may, for example, be one or more ARM, Intel x86, PowerPC processors or the like.

[0045] The memory **320** allows data to be stored and retrieved. The memory **320** may include, for example, random access memory, read-only memory, and persistent storage. Persistent storage may be, for example, flash memory, a solid-state drive, or the like. Read-only memory and persistent storage are non-transitory computer-readable storage mediums. A computer-readable medium may be organized using a file system such as may be administered by an operating system governing overall operation of the example computer system **300**.

[0046] The communications module **330** allows the example computer system **300** to communicate with other computer or computing devices and/or various communications networks. For example, the communications module **330** may allow the example computer system **300** to send or receive communications signals to/from the agent devices **110**, **150**, **160** over the network **140**. Communications signals may be sent or received according to one or more protocols or according to one or more standards. For example, the communications module **330** may allow the example computing system **300** to communicate via a cellular data network, such as for example, according to one or more standards such as, for example, Global System for Mobile Communications (GSM), Code Division Multiple Access (CDMA), Evolution Data Optimized (EVDO), Long-term Evolution (LTE) or the like. Additionally or alternatively, the communications module **330** may allow the example computing system **300** to communicate using near-field communication (NFC), via Wi-Fi™, using Bluetooth™ or via some combination of one or more networks or protocols. In some embodiments, all or a portion of the communications module **330** may be integrated into a component of the example computing system **300**. For example, the communications module **330** may be integrated into a communications chipset. In some embodiments, the communications module **330** may be omitted such as, for example, if sending and receiving communications is not required in a particular application.

[0047] The storage module **340** allows the example computing system **300** to store and retrieve data. In some embodiments, the storage module **340** may be formed as a part of the memory **320** and/or may be used to access all or a portion of the memory **320**. Additionally or alternatively, the storage module **340** may be used to store and retrieve data from persisted storage other than the persisted storage (if any) accessible via the memory **320**. In some embodiments, the storage module **340** may be used to store and retrieve data in a database. A database may be stored in persisted storage. Additionally or alternatively, the storage module **340** may access data stored remotely such as the database **130**, for example, as may be accessed using a local area network (LAN), wide area network (WAN), personal area network (PAN), and/or a storage area network (SAN). In some embodiments, the storage module **340** may access data stored remotely using the communications module **330**. In some embodiments, the storage module **340** may be omitted and its function may be performed by the memory **320** and/or by the at least one processor **310** in concert with the communications module **330** such as, for example, if data is stored remotely. The storage module may also be referred to as a data store.

[0048] Software comprising instructions is executed by the at least one processor **310** from a computer-readable medium. For example, software may be loaded into random-access memory from persistent storage of the memory **320**. Additionally or alternatively, instructions may be executed by the at least one processor **310** directly from read-only memory of the memory **320**.

[0049] FIG. **4** depicts a simplified organization of software components stored in the memory **320** of the example computing system **300** (FIG. **3**). As illustrated, these software components include an operating system **400** and an application **410**.

[0050] The operating system **400** is software. The operating system **400** allows the application **410** to access the at least one processor **310**, the memory **320**, and the communications module **330** of the example computing system **300** (FIG. **3**). The operating system **400** may be, for example, Google™ Android™, Apple™ iOS™, UNIX™, Linux™, Microsoft™ Windows™, Apple OSX™ or the like.

[0051] The application **410** adapts the example computing system **300**, in combination with the operating system **400**, to operate as a device performing a particular function. For example, the application **410** may cooperate with the operating system **400** to adapt a suitable embodiment of the example computing system **300** to operate as the server computing system **120** and/or the agent devices **110**, **150**, **160** (FIG. **1**).

[0052] While a single application **410** is illustrated in FIG. **4**, in operation, the memory **320** may include more than one application **410** and different applications may perform different operations. For example, in at least some embodiments in which the example computing system **300** is functioning as the agent device **110**, **150**, **160**, the applications **410** may include an application for displaying a graphical user interface associated with sending an application programming interface request. The server computer system **120** may be configured to receive application programming interface requests and may perform operations to respond thereto.

[0053] FIG. **5** is a simplified schematic diagram showing components of a host platform **500** of the server computer system **120** in greater detail and the database **130**. The server computer system **120** may store computer-executable instructions in the memory **320**, which may be executed by a processing unit such as the processor **310**, to implement one or more embodiments disclosed herein. The depicted example embodiments are directed to the server computer system **120** that hosts the host platform **500** that uses trained machine learning (ML) models, including generative artificial intelligence (GenAI), to enhance operations of a contact center or call center. The host platform **500** may receive input from a user, such as a call center agent and/or a caller through agent devices **110**, **150**, **160** where the input may be a text input, a document, or speech that is then converted into text and the like, and respond with an answer to the input. Here, one or more trained ML models, including a generative artificial intelligence (GenAI) model, may receive the input from the user and generate an answer based on its/their training. The response may include a text-based description, an image, a combination thereof, call routing, management of call queues, and the like.

[0054] The memory **320** of the server computer system **120** may store instructions for implementing software applications hosted by the host platform **500**, including an application interface **502**, a call monitoring module **510**, a response module **520**, trained models **530**, and data storage containing call center database **540**.

[0055] In the depicted example, the host platform **500** may be a cloud platform, web server, etc., that hosts software applications and other software programs that are hosted and made available on the Internet to the agent devices **110**, **150**, **160**. The software may be accessed via a URL, mobile application, etc. In other examples, the call monitoring module **510** and the response module **520** may reside in the memory **320** of the agent devices **110**, **150**, **160** while the trained models **530** and the call center database **540** may reside in the memory **320** of the server computer system **120**. Other variations are possible.

[0056] In any case, the application interface **502** may act as a software intermediary that allows an application executing on the agent device **110**, **150**, **160** to communicate with an application executing on the server computer system **120**. The application interface **502** may allow the agent device **110** to request data and may enable the server computer system **120** to obtain and provide the requested data to the agent device **110**.

[0057] The application interface **502** may be configured to receive application programming interface requests that define parameters. The application interface **502** may perform operations to obtain data to fulfill the application programming interface requests.

[0058] In one or more embodiments, the application interface **502** may include a representational state transfer (REST) application programming interface. The REST application programming interface may utilize Hypertext Transfer Protocol (HTTP) methods (e.g. GET, POST) to receive and respond to application programming interface requests. The REST application programming interface may obtain data according to application programming interface requests and may return fixed data sets as a response to the application programming interface requests.

[0059] In one or more embodiments, the application interface **502** may include a GraphQL application programming interface. The GraphQL application programming interface may be hierarchical. The GraphQL application programming interface may obtain data according to application programming interface requests without under fetching or over fetching data.

[0060] The application interface **502** may include both the REST application programming interface and the GraphQL schemas and may perform operations to select one of the REST and GraphQl application programming interfaces. In one or more embodiments, the server computer system **120** may receive an application programming interface request in a format compliant with one of the application programming interface schemas and may translate the request into another format.

[0061] The call monitoring module **510** comprises instructions to the processor **310** to monitor calls placed or directed to the agent device **110**, **150**, **160** and to identify one or more issues of the call in real-time during the call. To that end, the call monitoring module **510** may comprise a transcriber **512** and an issue identifier **514**. The transcriber **512** may be a speech-to-text application as known in the art that is configured to automatically convert the verbal/audio content of the call into a text format during the call. The transcriber **512** may be a software tool and/or a machine learning model trained to convert audio into a transcript in real-time.

[0062] The issue identifier **514** may be, or may comprise, another machine learning model that has been trained to identify one or more caller issues from the call. To that end, the issue identifier **514** may be trained to identify the caller issue from the audio portion of the call or from the transcript of the call (as transcribed by the transcriber **512**). The issue identifier **514** may be, for example, a trained neural network, a trained deep neural network (DNN), or a trained convolutional neural network (CNN). The issue identifier **514** may have been trained using a training dataset of audio calls and/or transcripts that have been labelled with ground-truth caller issue keywords. In that manner, the issue identifier

514 may be a large language model (LLM) (discussed below) that uses natural language processing techniques to extract the one or more caller issues from the call. These models may be stored in the memory **320** of the server computer system **120** as the trained models **530**, or may be stored and accessed remotely (not shown).

[0063] While the machine trained models described above may be specifically trained to transcribe an audio call and/or identify one or more caller issues from the call, in other applications, those (and other) functions may be performed by a foundational model, such as a refined or trained generative artificial intelligence (GenAI) model.

[0064] The response module **520** comprises instructions to the processor **310** to generate a response to be implemented by the call center in view of the one or more caller issues identified by the call monitoring module **510**. To that end, the response module **520** may be or may use a generative artificial intelligence (GenAI) model to generate a response. In the example embodiments, the host platform **500** may include one or more trained machine learning models, including GenAI model **532**, which is capable of receiving prompts and generating call center responses to the prompts. A prompt is typically understood to be a natural language input that includes instructions to the LLM/GenAI model to generate a desired output. The GenAI model **532** may be held by the host platform **500** within a model repository as one of the trained models **530**, or held and accessed remotely from a cloud.

[0065] According to various embodiments, the GenAI model **532** may be a large language model (LLM), such as a multimodal large language model. As another example, the GenAI model may be a transformer neural network ("transformer") or the like. A language model may use a neural network (typically a DNN) to perform natural language processing (NLP) tasks such as language translation, image captioning, grammatical error correction and natural language generation, among others. A language model may be trained to learn parameters in order to model how words relate to each other in a textual sequence, based on probabilities. A language model may contain hundreds of thousands of learned parameters or in the case of a large language model (LLM) may contain millions or billions of learned parameters or more. In that manner, the GenAI model can learn the patterns and structure of their input training data and then generate new content that has similar characteristics.

[0066] FIG. **6** schematically illustrates a process **600** of training the parameters of the GenAI model **532** according to example embodiments. However, it should be appreciated that the process **600** is also applicable to other types of models, such as other machine learning models, AI models, and the like. Referring to FIG. **6**, a host platform **610** may host an IDE **620** (integrated development environment) where GenAI models, machine learning models, AI models, and the like may be developed, trained, retrained, and the like. In this example, the IDE **620** may include a software application with a user interface accessible by a user device over a network or through a local connection.

[0067] For example, the IDE **620** may be embodied as a web application that can be accessed at a network address, URL, etc., by a device. As another example, the IDE **620** may be locally or remotely installed on a computing device used by a user.

[0068] The IDE **620** may be used to design a model (via a user interface of the IDE), such as a generative artificial intelligence model that can receive audio and/or text as input and generate custom responses for the call center, etc. The model can then be executed/trained based on training data established via the user interface. During training, the GenAI model **532** may be executed on training data via an AI engine **630** of the host platform **610**.

[0069] The GenAI model **532** may be trained to understand playbooks of call centers, natural language conversations, and the like based on a large corpus of documentation. The training data may be provided from a training data store such as an internal database **640**, which may include training samples from the web, from customers, and the like. Additionally or alternatively, the training data may be pulled from one or more external databases **650** such as publicly available sites, etc.

[0070] In some implementations, the GenAI model **532** may be trained with web pages, various playbooks and operation manuals of call centers, audio and/or transcripts of past calls to call centers, documentation, and other data about the operation of call centers. The GenAI model **532** may also be trained with content gathered by web page scrapers or crawlers from website pages comprising call center best practices, recommendations, and frequently asked question (FAQ) resources, and/or to fetch them from repositories or storage. The GenAI model **532** may further be trained with data from forum discussions, Q&A platforms, user reviews, customer satisfaction surveys, and other data sources that provide insight into the call center.

[0071] In some embodiments, the payload of data may be in a format that is not capable of being input to the GenAI model **532** nor read by a computer processor. For example, the payload of data may be in text format, image format, audio format, and the like. In response, the AI engine **630** may convert the payload of data into a format that is readable by the GenAI model **532**, such as a vector or other encoding. The vector may then be input to the GenAI model **532**.

[0072] The AI engine **630** may iteratively retrieve additional training data sets from the internal and external databases **640**, **650** and iteratively input the additional training data sets into the GenAI model **532** during the execution of the model to continue to train the model. The AI engine **630** may continue the process until it receives instructions to terminate, which may be based on a number of iterations (training loops), total time elapsed during the training process, etc.

[0073] When the GenAI model **532** is sufficiently trained, it may be stored within a model repository **660** as one of the trained models **530** via the IDE **620** or the like.

[0074] The IDE **620** may also be used to retrain the GenAI model **532** after the model has been deployed. Here, the training process may use executional results that have already been generated or output by the GenAI model **532** in a live environment (including any agent or caller feedback, etc.) to retrain the GenAI model **532**. For example, responses that are generated by the GenAI model **532** and the caller/agent/user feedback of the responses may be used to retrain the model to further enhance the responses that are generated for all users. The feedback may include indications of whether the generated responses match, are liked, and/or are relevant to the caller issue, and, if not, what aspects of the response are incorrect. This feedback data

may be captured and stored within a feedback data store **670** or other data store within the live environment and can be subsequently used to retrain the GenAI model **532**.

[0075] Returning to FIG. **5**, the response module **520** comprises instructions to the processor **310** to execute the trained GenAI model **532** to generate a response to the one or more caller issues (identified by the call monitoring module **510**) to be implemented by the call center in real-time during the call. The response may be to summarize the relevant information within a repository of the call center's playbook, provide answers to queries customized to the caller, identify solutions to the caller issue within past call transcripts, manage the call center queue, route calls to a particular agent, and other similar call center features. In some implementations, the desired output for the GenAI model **532** may be pre-set or predetermined. Alternatively, the response module **520** may also comprise instructions to the processor **310** to receive an indication from the agent device **110** regarding what the agent's desired output from the GenAI model **532** is or may be.

[0076] The host platform **500** may further host a call center database **540** containing the call center's current playbook **550** and call logs **560**. The call logs **560** may include transcripts **562** of prior calls made to the call center and metadata **564** associated with the prior calls and transcripts **562**. The prior call transcripts **562** may be verbatim or shorthand transcription of the conversation between the caller and the agent, and may include other call content data, including the caller issues, whether the issue(s) were resolved, how the issue(s) was resolved etc. The metadata **564** may include the date and time of call, the call duration, the wait time, the identity of the caller, the identity of the agent(s), and customer satisfaction feedback etc.

[0077] The host platform **500** may further host the database **130** or the database **130** may be provided remotely from the server computer system **120** (depicted in FIG. **5**). As noted above, in some embodiments, the server computer system **120** may be associated with a call center of a financial institution and, to that end, may maintain records of customer financial accounts and associated financial data in the database **130** as account information **570**.

[0078] According to various embodiments, to generate a response, the GenAI model **532** may be executed using custom-defined prompts that may be designed to first identify the caller issue (via the call monitoring module **510**), and then to generate a customized response related to the caller issue as noted above in view of the call center playbook, past calls, account information, or the like. In other embodiments, a separate prompt may be designed to perform each of those functions. In the depicted embodiment, the response module **520** may comprise a prompt generator **522** for generating prompts to the GenAI model **532**, such as to generate the customized response related to the caller issue in view of the call center playbook, past calls, account information etc.

[0079] Prompt engineering is the process of structing sentences (prompts) so that they are understood by the GenAI model. Part of the prompting process may include delays/waiting times that are intentionally included within the script such that the model has time to think/understand the input data.

[0080] The prompt generator **522** may comprise instructions to the processor **310** to generate a prompt to the GenAI model **532** (with the identified caller issue and with reference to the playbook **550**) to generate a number of different outputs. In some implementations, the desired output from the GenAI model **532** may be pre-set or predetermined. Alternatively, a signal regarding the desired output from the GenAI model **532** may be previously received by the processor **310** from the agent device **110**. In either case, the prompt may also be generated with the pre-set or indicated desired output specified. The response module **520** further comprises instructions to the processor **310** to implement the response (based on the generated output) during the call. For example, implementation of the response may include providing the output from the GenAI model **532** on a user interface of the agent device **110**, such as on the display **210**.

[0081] In some implementations, the prompt generator **522** may comprise instructions to the processor **310** to generate a prompt for generating a summary of a portion of the playbook **550** related to the caller issue. The summary generated by the GenAI model **532** may then be implemented when the response module **520** instructs the processor **310** to provide the summary output to the requesting agent via the user interface, such as to be displayed on the display **210** of the agent device **110** in real-time during the call. This saves on processing power, as the agent no longer has to search through the playbook for the relevant portions, and reduces response time as the agent no longer has to search for and review the content before responding to the caller.

[0082] In other implementations, the prompt generator **522** may comprise instructions to the processor **310** to receive an authorization regarding an identity of the caller from the agent device **110**. To that end, the memory of the agent device **110** may comprise instructions to implement an authorization protocol with the caller during the call. The prompt generator **522** may comprise instructions to the processor **310** to generate a prompt that further incorporates the account information **570** or includes reference to the account information **570** (such as may be stored in the database **130** of the server computer system **120**) associated with the identity of the caller.

[0083] In some implementations, when the caller issue is in a form of a question, the generated prompt may be for generating a reply based on the playbook **550** customized according to the account information **570**. The reply generated by the GenAI model **532** may then be implemented when the response module **520** instructs the processor **310** to provide the reply output to the requesting agent via the user interface, such as to be displayed on the display **210** of the agent device **110** in real-time during the call. This saves on processing power, as the agent no longer has to search through the caller's account information, and reduces response time as the agent no longer has to search for and review the caller's account information before responding to the caller.

[0084] As noted above, the host platform **500** may host the call center database **540** containing the call center's current playbook **550** and call logs **560** comprising transcripts **562** of prior calls and metadata **564** associated with the prior call transcripts **562**. In other implementations, the prompt generator **522** may comprise instructions to the processor **310** to generate a prompt for identifying the caller issue (as identified by the monitoring module **510**) in one of the prior call transcripts **562**, and for identifying a solution to the caller issue in the identified prior call transcript **562**. The solution generated by the GenAI model **532** may then be imple-

mented when the response module **520** instructs the processor **310** to provide the solution output to the requesting agent via the user interface, such as to be displayed on the display **210** of the agent device **110** in real-time during the call. This saves on processing power, as the agent does not have to search through all of the prior call transcripts **562** for the caller issue and associated solution, and reduces response time.

[0085] If the GenAI model **532** is an LLM, the prompts and the summary and the reply outputs may be in a natural language format.

[0086] The response module **520** may further have instructions to the processor **310** to ask for, and receive, feedback from the agent about the output generated by the GenAI model **532** (such as the summary, the reply, the solution etc.) via the user interface. As noted above, this feedback data may be captured and stored within the feedback data store **670** or other data store within the live environment and can be subsequently used to retrain the GenAI model **532**. The retrained GenAI model may be stored in the memory **320** with the other trained models **530**.

[0087] As noted above, the call logs **560** of the call center database **540** may include transcripts **562** of prior calls and metadata **564** associated with the prior call transcripts **562**. The metadata **564** may include the date and time of call, the call duration, the wait time, the identity of the caller, the identity of the agent(s), and customer satisfaction feedback etc. In view of the call logs **560**, the prompt generator **522** may also comprise instructions to the processor **310** to generate a prompt to the GenAI model **532** to generate other outputs that may be implemented by the processor **310** as the response in the following ways.

[0088] In some implementations, the prompt generator **522** may comprise instructions to the processor **310** to generate a prompt that further includes a reference to the call logs **560**. The prompt may be generated for identifying the caller issue (as identified by the monitoring module **510**) in one of the prior call transcripts **562**, and for identifying the call duration of the prior call of the identified prior call transcripts **562**. Thus, the output from the GenAI model **532** may comprise an expected call duration of the current call based on the call duration of the prior call of the identified prior call transcripts. The response module **520** may have further instructions to the processor **310** to implement the response by placing the current call in a call queue according to the expected call duration. In some implementations, the prompt generator **522** may comprise instructions to the processor **310** to receive authorization regarding the identity of the caller from the agent device **110**. The prompt generator **522** may then further comprise instructions to the processor **310** to generate the prompt which includes reference to the account information **570** (such as may be stored in the database **130** of the server computer system **120**) associated with the identity of the caller. The expected call duration generated by the GenAI model **532** may then be further based on the account information **570** of the caller. In that manner, a call queue of the call center may be managed based on the expected call durations of different live calls.

[0089] In other implementations, the response module **520** may comprise instructions to the processor **310** to implement the response by routing the call to a particular agent based on the output from the GenAI model **532**. To that end, the prompt generator **522** may comprise instructions to the processor **310** to generate a prompt that further includes a reference to the call logs **560**. The prompt may be generated for identifying the current caller issue (as identified by the monitoring module **510**) in one of the prior call transcripts **562**, and for identifying the agent associated with the prior call of the identified prior call transcript **562**. This helps to identify an agent who has dealt with the current caller issue in the past, as evidenced by the prior call transcript **562**. Thus, the output from the GenAI model **532** may comprise the identification of the agent associated with the prior call of the identified prior call transcript. The response module **520** may have further instructions to the processor **310** to implement the response by routing the call to the identified agent. This may help to save on processing power, as a more inexperienced agent may not have to search through all of the prior call transcripts **562** and/or the playbook **550** in order to find the caller issue and associated solution, or may help to save on processing power as this may help to prevent an inexperienced agent from having to try potentially ineffective solutions before arriving at an effective one. This may also help to reduce response time.

[0090] The response module **520** may further have instructions to the processor **310** to ask for, and receive, feedback from the identified agent about the routing via the user interface. As noted above, this feedback data may be captured and stored within the feedback data store **670** or other data store within the live environment and can be subsequently used to retrain the GenAI model **532**. The retrained GenAI model may be stored in the memory **320** with the other trained models **530**.

[0091] Reference will now be made to FIG. **7**, which shows, in flowchart form, an example method **700** of enhancing call center features based on the call center's playbook according to example embodiments. The method **700** may be implemented by way of suitably programmed processor-executable instructions stored in memory that, when executed, cause a computing device to carry out the described functions as described above. As other examples, the method **700** may be performed by another computing system, a software application, a server, a cloud platform, a combination of systems, and the like.

[0092] At operation **702**, the method **700** may include training the GenAI model. The GenAI model may be a new or existing foundational model refined or trained to understand playbooks of call centers, natural language conversations, and the like based on a large corpus of documentation. The training data may be provided from a training data store, which may include training samples from the web, from customers, and the like. Additionally or alternatively, the training data may be pulled from one or more external databases, such as publicly available sites, etc. The GenAI model may be trained with web pages, various playbooks and operation manuals of call centers, audio and/or transcripts of past calls to call centers, documentation, and other data about the operation of call centers. The GenAI model may also be trained with content gathered by web page scrapers or crawlers from website pages comprising call center best practices, recommendations, and frequently asked question (FAQ) resources, and/or to fetch them from repositories or storage. The GenAI model may further be trained with data from forum discussions, Q&A platforms, user reviews, customer satisfaction surveys, and other data sources that provide insight into the operations of the call center.

[0093] The customized GenAI model may then be stored (in memory) at operation **704**. Other trained machine learning models may also be stored in memory, including a speech-to-text model and/or a machine learning model that has been trained to identify one or more caller issues from the call (such as the issue identifier discussed above). This model may be a trained neural network, a trained deep neural network (DNN), or a trained convolutional neural network (CNN) trained on training datasets of audio calls and/or transcripts that have been labelled with ground-truth caller issue keywords. In that manner, it may be a large language model that uses natural language processing (NLP) techniques to extract one or more caller issues from the call.

[0094] At operation **706**, the playbook of the call center may also be stored in memory or a database, along with, optionally, prior versions of the playbook, call transcriptions or audio files of prior calls to the call center (at operation **708**) and associated metadata (at operation **710**). The transcripts of prior calls made to the call center may be verbatim or shorthand transcription of the conversation between the caller and agent, and may include other call content data, including the caller issues, whether the issue(s) were resolved, how the issue(s) was resolved etc. The metadata may include the date and time of call, the call duration, the wait time, the identity of the caller, the identity of the agent(s), and customer satisfaction feedback etc.

[0095] At operation **712**, a call between the caller and the call center agent is monitored in real-time, such as may be performed by the call monitoring module described above. The call may be monitored and recorded as an audio file. Alternatively or additionally, a transcript of the call may be obtained at operation **714** in real-time during the call with the caller. The transcript may be obtained using a speech-to-text tool, such as a transcriber (at operation **716**). The transcriber may be a software application, such as the speech-to-text model stored at operation **704**. Alternatively, the transcriber may be stored remotely and accessed as needed.

[0096] At operation **718**, the method **700** further includes identifying, from the call or transcript, a caller issue in real-time using the same or another trained machine learning model, such as the issue identifier discussed above. This machine learning model may have been trained using a training dataset of audio calls and/or transcripts that have been labelled with ground-truth caller issue keywords. In that manner, the issue identifier may be an LLM that uses natural language processing techniques to extract the one or more caller issues from the call. The issue identifier used may have been stored locally at operation **704**. Alternatively, the issue identifier may be stored remotely and accessed as needed.

[0097] At operation **720**, a response to the caller issue is obtained based on execution of the GenAI model and the playbook stored in memory to generate an output. In order to execute the GenAI model, a prompt may be generated based on the identified caller issue and the playbook at operation **722**, such as by the prompt generator discussed above. A prompt is typically a natural language input that includes instructions to the LLM/GenAI model to generate a desired output.

[0098] The desired output may be pre-set or predetermined. Alternatively, the method **700** may also include receiving an indication from the agent device regarding the agent's desired output from the GenAI model **532** prior to generating the prompt.

[0099] For example, if the desired output is a summary of a portion of the playbook related to the caller issue, the prompt may be engineered and generated for that purpose, and the output would include a summary of the playbook related to the caller issue.

[0100] In other implementations, if the desired output involves output customized to a caller's personal account information, at operation **724**, the prompt may be generated to incorporate account data or information associated with the caller. For example, if the call center was associated with a financial institution, the account data may be the banking or financial information of the caller's account with the financial institution. In some implementations, the account information may be retrieved from memory and incorporated in the prompt. Alternatively, the prompt may be generated to link or refer to the account information stored remotely. Prior to operation **724**, authorization regarding an identity of the caller may be received prior to the caller's account information being incorporated into, or referred to in, the prompt. For example, if the caller issue is in a form of a question, the prompt may generated to output a reply based on the playbook customized according to the caller's account information.

[0101] If the desired output relates to content from prior calls to the call center, at operation **726**, the prompt may be generated to identify or reference call transcriptions or audio files of prior calls to the call center (saved at operation **708**). For example, if the desired output is a solution to the current caller issue that was implemented in a past call, the prompt may be engineered and generated to identify the current caller issue in one of the prior call transcripts, and to identify the associated solution. The output would then include the solution from the prior call.

[0102] If the desired output relates to parameters or metadata associated with prior calls to the call center, at operation **726**, the prompt may also be generated to identify or reference metadata associated with the transcripts or audio files of prior calls to the call center (saved at operation **710**). For example, if the desired output is an expected call duration of the current call, the prompt may be engineered and generated to identify the current caller issue in one of the prior call transcripts (or audio files), and to identify the call duration of the prior call from the associated metadata. The output would then include the expected call duration of the current call. If authorization regarding the identity of the caller was received, the caller's account information may also be incorporated into, or be referred to in, the prompt. Thus, the output may then include the expected call duration of the current call for the particular caller.

[0103] In other implementations, if the desired output is an identity of an agent who has handled the current caller issue before at the call center, the prompt may be engineered and generated to identify the current caller issue in one of the prior call transcripts, and to identify the agent associated with the prior call. Thus, the output from the GenAI model **532** may comprise the identification of the agent associated with the prior call of the identified prior call transcript. The output would then include the identity of the agent who has handled the current caller issue before at the call center.

[0104] The response to the caller issue, then, may be based on the above-described output from the GenAI model, and the response implemented at operation **728** in real-time during the call.

[0105] Implementation of the response would depend on the output generated by the GenAI model. For example, if the output is a summary of the playbook related to the caller issue, an answer to a question customized to the caller's account information, or a solution to the caller issue used in a prior call, at operation **730**, the implementation may involve displaying the output to the agent on a user interface during the call. If the output is an expected call duration for the current call, at operation **732**, the implementation may involve placing the current call in a call queue according to the expected call duration. In that manner, a call queue of the call center may be managed based on the expected call durations of different live calls. If the output is an identity of an agent who has handled the current caller issue before, at operation **734**, the implementation may involve routing the current call to the identified agent.

[0106] At operation **736**, feedback may be received from the agent about the output generated by the GenAI model (such as the summary, the reply, the solution, the routing etc.) via the user interface. This feedback data may be captured and stored in memory or other data store and can be subsequently used to retrain the GenAI model at operation **702**. The retrained GenAI model may be stored in the memory at operation **704** with the other trained models.

[0107] The presently described systems and methods may help an agent respond more accurately and consistently to caller issues in relation to the call center's playbook, without extensive and time-consuming searching of the playbook and prior call transcripts or resorting to imperfect recollection. The presently described systems and methods may also help the call center manage live calls by more effectively ordering call queues and routing callers to an experienced or appropriate agent, who can more quickly resolve the caller issue. This may help to save on processing power, as it may help prevent overlap of the same searching of the playbook and/or prior call transcripts when multiple agents encounter the same caller issue.

[0108] The methods described herein may be modified and/or operations of such methods combined to provide other methods.

[0109] Example embodiments of the present application are not limited to any particular operating system, system architecture, mobile device architecture, server architecture, or computer programming language.

[0110] It will be understood that the applications, modules, routines, processes, threads, or other software components implementing the described method/process may be realized using standard computer programming techniques and languages. The present application is not limited to particular processors, computer languages, computer programming conventions, data structures, or other such implementation details. Those skilled in the art will recognize that the described processes may be implemented as a part of computer-executable code stored in volatile or non-volatile memory, as part of an application-specific integrated chip (ASIC), etc.

[0111] As noted, certain adaptations and modifications of the described embodiments can be made. Therefore, the herein discussed embodiments are considered to be illustrative and not restrictive.

What is claimed is:

1. A server computer system, comprising:
a processor;
a communications module coupled to the processor; and
a memory coupled to the processor, the memory storing a playbook of a call center and instructions that, when executed, configure the processor to:
monitor a call in real-time during the call with a caller;
identify, from the call, a caller issue in real-time using a trained machine learning model;
obtain a response to the caller issue based on execution of a generative artificial intelligence (GenAI) model and the playbook stored in the memory; and
implement the response during the call.

2. The system of claim **1**, wherein the instructions, when executed, further configure the processor to: obtain a transcript of the call in real-time during the call, and identify the caller issue from the transcript of the call.

3. The system of claim **2**, further comprising a transcriber coupled to the processor, wherein the transcript of the call is obtained by the transcriber.

4. The system of claim **1**, wherein the instructions, when executed, further configure the processor to train the GenAI model with other call center playbooks prior to the call.

5. The system of claim **3**, wherein the GenAI model is a large language model (LLM).

6. The system of claim **5**, wherein the instructions, when executed, further configure the processor to:
obtain the response by:
generating a prompt to the LLM, the prompt including the caller issue with reference to the playbook; and
obtain an output from the LLM responsive to the prompt; and
implement the response by providing the output via a user interface.

7. The system of claim **6**, wherein the output comprises a summary of a portion of the playbook related to the caller issue, and the prompt to the LLM is for generating the summary as the output.

8. The system of claim **6**, wherein the instructions, when executed, further configure the processor to receive an authorization regarding an identity of the caller from an agent device.

9. The system of claim **8**, wherein the prompt further includes reference to account information associated with the identity of the caller.

10. The system of claim **9**, wherein the caller issue is a form of a question, the output comprises a reply to the question, and the prompt to the LLM is for generating the reply based on the playbook customized according to the account information.

11. The system of claim **6**, wherein the memory further stores prior call transcripts of the call center, wherein the prompt to the LLM is for identifying:
the caller issue in one of the prior call transcripts, and
a solution to the caller issue in the one of the prior call transcripts; and
wherein the output comprises the solution.

12. The system of claim **11**, wherein the instructions, when executed, further configure the processor to receive feedback about the solution via the user interface, retrain the GenAI model based on execution of the GenAI model on the caller issue, the solution, and the feedback, and store the retrained GenAI model in the memory.

13. The system of claim 2, wherein the memory further stores call log data of the call center, the call log data comprising prior call transcripts of prior calls and metadata associated with the prior calls, wherein the instructions, when executed, further configure the processor to:

obtain the response by:

generating a prompt to the GenAI model, the prompt including the caller issue with reference to the playbook and the call log data, and

obtain an output from the GenAI model responsive to the prompt; and

implement the response based on the output.

14. The system of claim 13, wherein the metadata comprises a call duration of each of the prior calls, wherein the prompt to the GenAI model is for identifying:

the caller issue in one of the prior call transcripts, and

the call duration of the prior call of the one of the prior call transcripts;

wherein the output comprises an expected call duration of the call based on the call duration of the prior call of the one of the prior call transcripts; and

wherein the instructions, when executed, further configure the processor to implement the response by placing the call in a call queue according to the expected call duration.

15. The system of claim 14, wherein the prompt further includes account information associated with an identity of the caller, and wherein the expected call duration is further based on the account information of the caller.

16. The system of claim 13, wherein the instructions, when executed, further configure the processor to implement the response by routing the call to a particular agent based on the output.

17. The system of claim 13, wherein the metadata comprises identification of an agent associated each of the prior calls, wherein the prompt to the GenAI model is for identifying:

the caller issue in one of the prior call transcripts, and

the agent associated with the prior call of the one of the prior call transcripts;

wherein the output comprises the identification of the agent associated with the prior call of the one of the prior call transcripts; and

wherein the instructions, when executed, further configure the processor to implement the response by routing the call to the identified agent.

18. The system of claim 17, wherein the instructions, when executed, further configure the processor to receive feedback about the routing from the identified agent, retrain the GenAI model based on execution of the GenAI model on the caller issue, the routing, and the feedback, and store the retrained GenAI model in the memory.

19. A method comprising:

storing a playbook of a call center in a memory;

monitoring a call in real-time during the call with a caller;

identifying, from the call, a caller issue in real-time using a trained machine learning model;

obtaining a response to the caller issue based on execution of a generative artificial intelligence (GenAI) model and the playbook stored in the memory; and

implementing the response during the call.

20. A computer-readable medium comprising instructions stored therein which, when executed by a processor, cause a computer to:

store a playbook of a call center in a memory;

monitor a call in real-time during the call with a caller;

identify, from the call, a caller issue in real-time using a trained machine learning model;

obtain a response to the caller issue based on execution of a generative artificial intelligence (GenAI) model and the playbook stored in the memory; and

implement the response during the call.

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